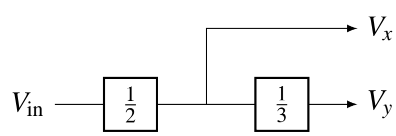
# EECS 16A Designing Information Devices and Systems I
## Discussion 12A

**1. Modular Circuit Buffer**

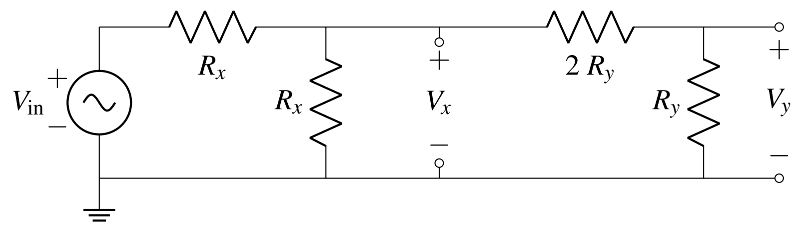
Let's try designing circuits that perform a set of mathematical operations using op-amps. While voltage dividers on their own cannot be combined without altering their behavior, op-amps can preserve their behavior when combined and thus are a perfect tool for modular circuit design. We would like to implement the block diagram shown below:




In other words, create a circuit with two outputs $V_x$ and $V_y$, where $V_x = \frac{1}{2}V_{in}$ and $V_y = \frac{1}{3}V_x = \frac{1}{6}V_{in}$.

(a) Draw two voltage dividers, one for each operation (the $1/2$ and $1/3$ scalings). What relationships hold for the resistor values for the $1/2$ divider, and for the resistor values for the $1/3$ divider?

(b) If you combine the voltage dividers, made in part (a), as shown by the block diagram (output of the $1/2$ voltage divider becomes the source for the $1/3$ voltage divider circuit), do they behave as we hope (meaning $V_{in} = 2V_x = 6V_y$)?

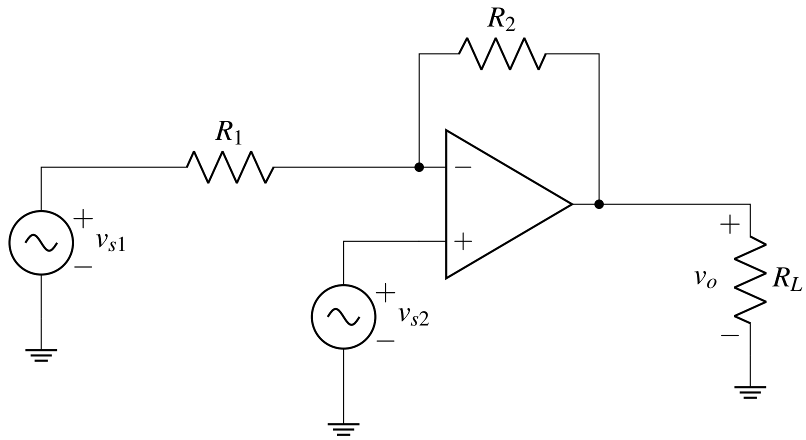


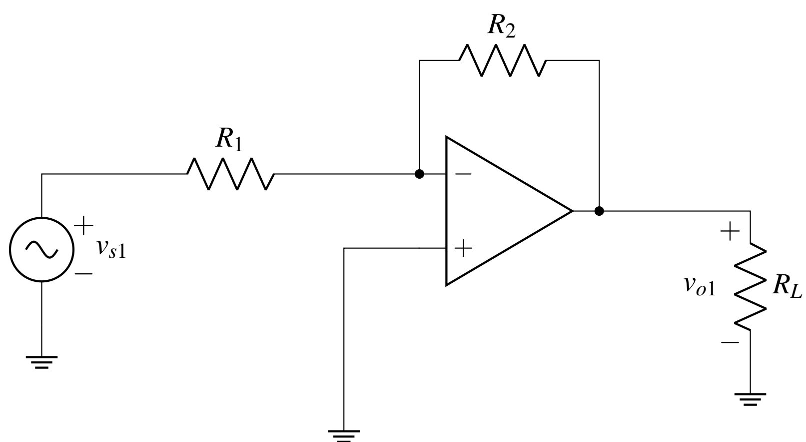
(c) Perhaps we could use an op-amp (in negative-feedback) to achieve our desired behavior.
Modify the implementation you tried in part (b) using a negative feedback op-amp in order to achieve the desired $V_x, V_y$ relations $V_x = \frac{V_{\text{in}}}{2}$ and $V_y = \frac{V_x}{3} = \frac{V_{\text{in}}}{6}$.
HINT: Place the op-amp in between the dividers such that the $V_x$ node is an input into the op-amp, while the source of the 2nd divider is the output of the op-amp!

**2. Amplifier with Multiple Inputs**

In this problem we will use superposition and the Golden Rules to find the output of the following op amp circuit with multiple inputs:

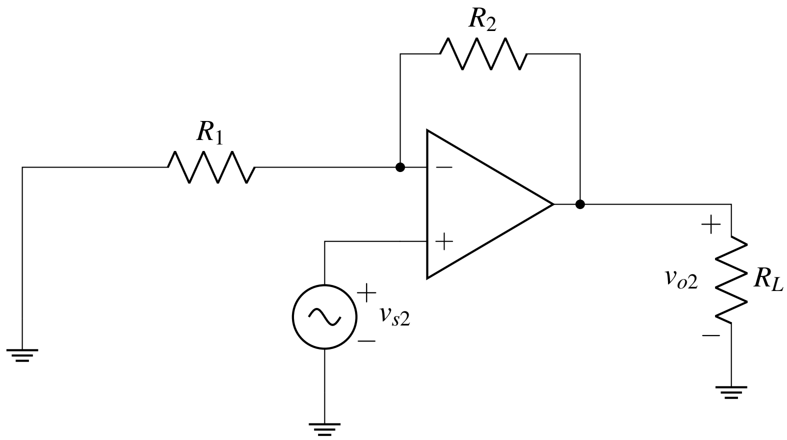



(a) First, let's turn off $v_{s2}$. Use the **Golden Rules** to find $v_{o1}$ for the circuit below.

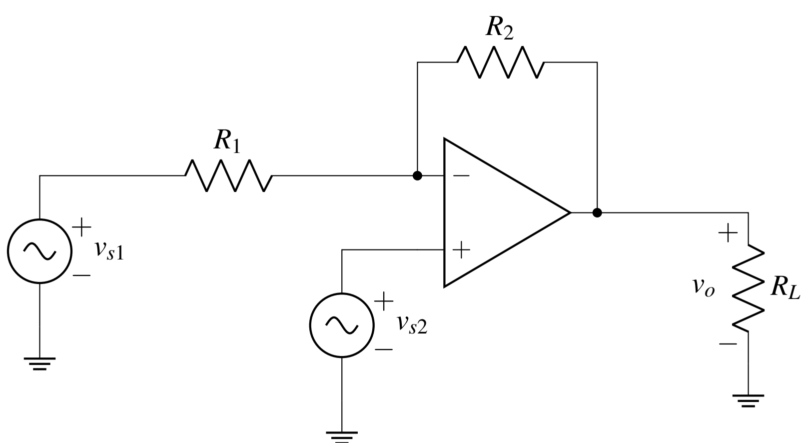


(b) Now let's turn off $v_{s1}$. Use the **Golden Rules** to find $v_{o2}$ for the circuit below.




(c) Use **superposition** to find the output voltage $v_o$ for the circuit shown below.



**3. Modular Op-Amp Circuits**

Let's design blocks that implement the following operations:

(a) Scale the input voltage so that: $V_{\text{out}} = +5\ V_{\text{in}}$.

(b) Scale and invert the input voltage so that: $V_{\text{out}} = -2\ V_{\text{in}}$.